

Lab # 13

Exception Handling

Lab Tasks:-

- Write a program that meets the following requirements:
- Creates an array with 10 randomly chosen integers.
 - Prompts the user to enter the index of the array, then displays the corresponding element value. If the specified index is out of bounds, display the message Out of Bounds. (ArrayIndexOutOfBoundsException)

Code:-

```
ArrayIndexDemo.java x KidsDivisionGame.java
1  import java.util.Random;
2  import java.util.Scanner;
3  public class ArrayIndexDemo {
4      public static void main(String[] args) {
5
6          int[] numbers = new int[10];
7          Random rand = new Random();
8
9          for (int i = 0; i < numbers.length; i++) {
10             numbers[i] = rand.nextInt( bound: 100);
11         }
12
13         Scanner scanner = new Scanner(System.in);
14
15         try {
16             System.out.print("Enter an index (0-9): ");
17             int index = scanner.nextInt();
18
19             System.out.println("Value at index " + index + ": " + numbers[index]);
20
21         } catch (ArrayIndexOutOfBoundsException e) {
22             System.out.println("Out of Bounds");
23         }
24     }
25 }
```

Output:-

```
"C:\Program Files\Eclipse Adoptium\
Enter an index (0-9): 7
Value at index 7: 68

Process finished with exit code 0
```

- Suppose you are developing a game for kids in which they are learning the division operation in math. Your game will take input from kids (2 integers) and then perform the division and displays the answer. Think and apply exception handling in this scenario. For e.g. Arithmetic Exception might occur here. Also suppose, this game has the limitation that it only performs division between integers so if it gets a decimal number as input, it throws an exception stating that the input is invalid, please give integer number etc.

Code:-

```
KidsDivisionGame.java
1  import java.util.Scanner;
2  import java.util.InputMismatchException;
3  public class KidsDivisionGame {
4      public static void main(String[] args) {
5
6          Scanner scanner = new Scanner(System.in);
7
8          try {
9              System.out.print("Enter the first integer: ");
10             int num1 = scanner.nextInt();
11
12             System.out.print("Enter the second integer: ");
13             int num2 = scanner.nextInt();
14
15             int result = num1 / num2;
16
17             System.out.println("Result: " + result);
18         } catch (ArithmeticException e) {
19             System.out.println("Error: Cannot divide by zero.");
20         } catch (InputMismatchException e) {
21             System.out.println("Invalid input! Please enter integer numbers only.");
22         } finally {
23             // Cleanup or final actions
24         }
25     }
26 }
27
28 }
```

Output:-

```
"C:\Program Files\Eclipse Adoptium
Enter the first integer: 25
Enter the second integer: 5
Result: 5

Process finished with exit code 0
```